BJIT

CandiVue AI

Software Requirement Specification

AI-assisted recruitment and interview platform

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Revision History

Doc.	Date	Author	Reviewer	Description of Revision
1.0	26/10/2026	Miftahul Islam		First complete version, written for the final evaluation. Covers features F1–F11 and the project decisions up to §4B-17.

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1. Introduction

1.1 Purpose

This document specifies the software requirements of CandiVue AI, an AI-assisted recruitment and interview platform built as the final project of the BJIT AI Trainee Developer Programme. It states what the system must do, the quality it must reach and the constraints it is built under, so that the evaluation panel, the trainer and any future developer share one agreed description of the product.

The requirements are derived from the issued Business Information Document (BID), the written trainer clarifications of 28 September 2026 and the project decisions recorded during development. Each requirement has an identifier, a measurable acceptance criterion and a reference to the test that verifies it, so the document can also be used as a checklist during the final demonstration.

1.2 Scope

CandiVue AI covers the hiring workflow of one organisation from the job description to an evidence-backed assessment of each interviewed candidate. It supports, but never replaces, the human decision.

Included

- Job profiles and job descriptions written with an AI assistant (the Hiring Assistant) that uses the organisation's own reference documents (RAG), with interviewer approval before anything is published.
- Publishing the approved job to a job board and collecting CVs through a Google Form, plus manual CV upload.
- Batch CV screening against the job's hard requirements, an explainable 0–100 score per candidate and a ranked shortlist (F1, F2).
- Interview scheduling with email invitations and calendar events, a private interview link per candidate and a one-time access code.
- A browser-based one-to-one video interview with a live, speaker-attributed and timestamped transcript (F3, F4).
- Candidate identification at check-in against the photo sent with the application, with consent (F7).
- Post-interview question-and-answer analysis, CV claim checks, an interview score and an interview quality review (F5, F8, F9, F10).
- A consolidated assessment report with interviewer review, export, candidate comparison and a final ranking (F6).
- Talent rediscovery: suggesting consenting past applicants for a new vacancy (F11, the project's innovation feature).
- Data protection functions: deletion of applicants and jobs, retention limits and an hourly clean-up.

Excluded

- Production or cloud hosting, CI/CD pipelines, load balancing and infrastructure-as-code. The system runs locally with Docker Compose.
- Enterprise authentication (SSO, LDAP). Only the Interviewer and Candidate roles exist.
- Integration with commercial ATS or HRIS products, payroll, offer management and onboarding. The selection email is a notification only and carries no offer details.
- Multi-tenancy, billing, native mobile apps and panel interviews (more than one interviewer).
- Voice recognition. F7 is implemented with face comparison only.
- Editing the transcript after the interview, and languages other than English.
- Training or fine-tuning AI models. Existing models and APIs are used.

Items that go beyond the BID (job-board posting, calendar integration, scoring of company-related answers, the selection email and face-only F7) are project decisions recorded in docs/requirement.md §4B and are subject to written trainer approval.

1.3 Intended Stakeholder

This document is written for the following readers:

Stakeholder	Interest in this document
Evaluation panel and trainer	Assess the product against the BID and this specification; confirm the open project decisions.
Trainee developer (author)	Builds, tests and demonstrates the system; keeps this document in line with the code.
Programme coordinators	Track deliverables, milestones and submission.
Interviewers (end users)	Create jobs, screen CVs, conduct interviews and review assessments.
Candidates (end users)	Apply through the form, join their interview and complete the identity check-in.
Future maintainers	Use the requirement IDs and test references to understand and change the system safely.

1.4 References

Reference	Location
Requirement Specification	Business Information Document (BID), AI-Powered Recruitment & Interview Intelligence Platform, v2, 34 pages
Trainer clarifications	Written trainer Q&A sheet, 28 September 2026
Requirements reference	docs/requirement.md (incl. project decisions §4B-1 to §4B-17)
Verification report	docs/VERIFICATION.md (security, privacy, performance and feature verification)
Architecture and workflows	docs/diagrams/ (high-level and detailed architecture, LangGraphs 1–5)
UI design	docs/design_handoff_candivue_ui/ (screen designs and interaction notes)
Setup and operation	README.md in the repository root
Source code and tests	BJIT GitHub repository candivue-ai-30225

1.5 Definitions, Acronyms, and Abbreviations

Term/Acronym	Definition
ATS match	A score for how closely the wording of a CV matches the job description, similar to an applicant tracking system.
BID	Business Information Document: the issued assignment brief.
Check-in	The candidate’s identity step before the interview: three camera pictures compared with the application photo.
Checkpoint	The saved state of a LangGraph workflow, used to resume after a failure or restart.
Embedding	A list of numbers that represents the meaning of a text; similar texts get similar numbers.
F1–F11	Feature identifiers from the BID (F1 CV screening … F11 innovation feature).
Gate	The hard-requirement check that excludes a candidate who lacks a mandatory requirement or the minimum experience.

Term/Acronym	Definition
HITL	Human in the loop: the workflow pauses until the interviewer approves or rejects.
Interviewer	A signed-in staff user who runs the hiring process. Has access to all hiring data.
JD	Job description.
JWT	JSON Web Token: the signed login token sent with each request.
LangGraph	A framework for multi-step AI workflows with explicit state, nodes and edges.
LLM	Large language model (Google Gemini in this project).
MCP	Model Context Protocol: the protocol used to reach the job board, email and calendar services.
OCR	Optical character recognition, used for scanned CVs.
Q&A unit	One interviewer question and the candidate's answer to it, as found in the transcript.
RAG	Retrieval-augmented generation: the AI is given relevant passages from the knowledge base.
STT	Speech-to-text.
Talent pool	Past applicants who agreed to be considered for other vacancies.

2. Overall Description

2.1 Overview

Product perspective

CandiVue AI is a new, self-contained web application. It is not part of an existing product. All of its own components run in Docker containers on one machine: the web frontend, the API backend, the database, a local AI model server, a video server, a transcription worker, a CV collector and three small integration servers. It relies on a few external services that the organisation already uses or can sign up for free: Google Gemini (AI model), Deepgram (speech-to-text), Gmail (email), Google Calendar, Google Forms and Drive (applications) and Blogger (job board).

Five AI workflows are modelled as LangGraph graphs: the Hiring Assistant (a tool-calling agent with approval pauses), CV Screening, Talent Rediscovery, Q&A Analysis and Final Assessment. The four workflows other than the assistant are fixed graphs, so the same input leads through the same steps and the scores stay repeatable.

Figure 1 shows the whole process in four stages, from preparing a job to the final decision, and who does each step: the interviewer, the candidate or the system on its own.

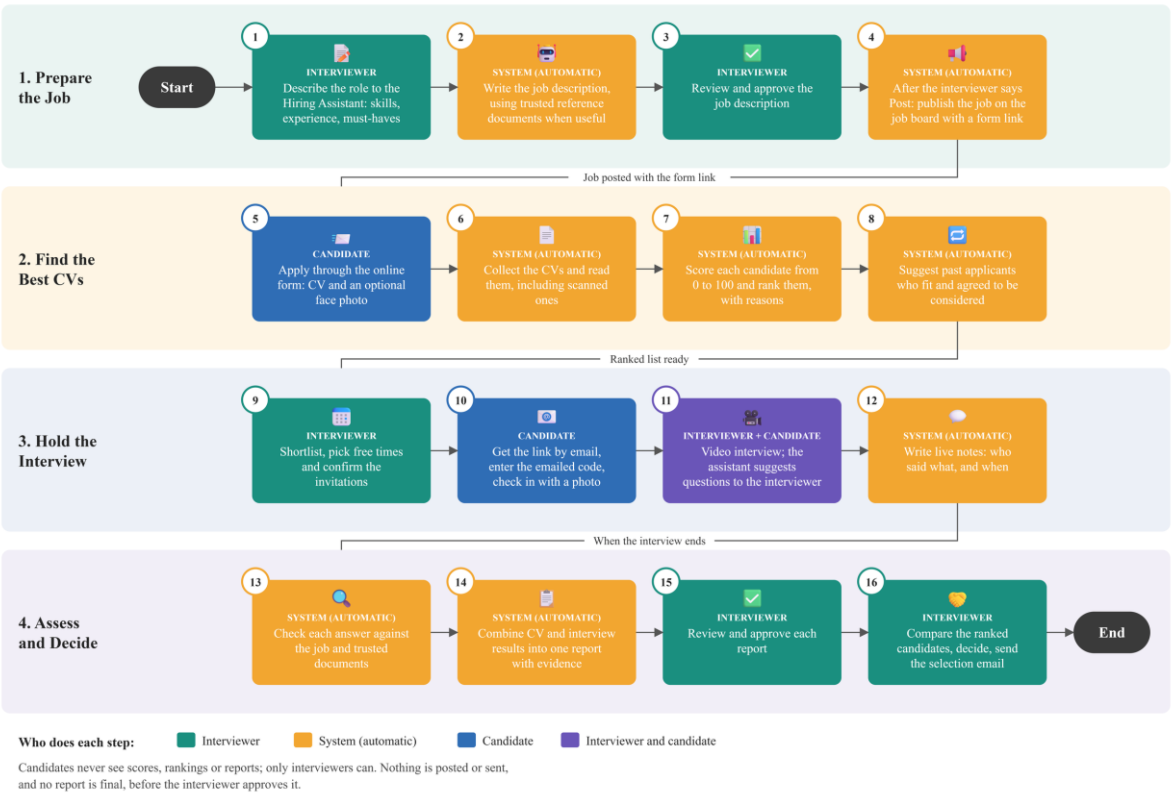


Figure 1: High Level System Diagram

Product functions

- Write and publish job descriptions with an AI assistant and the organisation’s own documents.
- Collect, read and screen CVs in batches, with an explained score and ranking.
- Suggest suitable past applicants from the talent pool.
- Schedule interviews, send invitations and manage the interviewer’s calendar.
- Run the video interview with a live transcript and an identity check-in.
- Analyse the interview and prepare an evidence-backed assessment for review.
- Rank approved candidates and send the selection email.

User characteristics

User class	Characteristics
Interviewer	Recruiter or hiring manager. Uses a desktop or laptop browser most of the time. Knows the job domain but is not expected to know AI or technical details. Makes every hiring decision; the system only advises.
Candidate	External applicant. Uses the private link from the invitation email on a laptop or phone with a camera and microphone. Needs no account. Sees only their own interview, never scores, rankings, transcripts or reports.

Constraints

- Python, FastAPI with Pydantic, LangGraph and Docker are mandatory (BID §8). The frontend technology is the developer's choice.
- The whole system must start on a clean machine with one documented command.
- Only the 20 sample CVs provided by the trainer or synthetic data may be used. No real personal data in the repository.
- The AI provider's free tier limits the number of requests per day, so the design keeps AI requests few.
- One developer, 20 working days, delivery by 27 October 2026.

Assumptions and dependencies

- The host machine runs Docker Desktop or Docker Engine with at least 8 GB of memory available to Docker and about 15 GB of free disk space for the images and local models.
- Internet access is available for Gemini, Deepgram and the Google services. Without it, screening and search still work with the local models, but AI writing, transcription and email do not.
- Interviews take place in a current version of Chrome or Edge with camera and microphone permissions.
- The organisation provides a Google account for the form, Drive, Calendar, Gmail and Blogger, and the keys are placed in the .env file.
- Trainer approval is obtained for the project decisions that go beyond the BID (job board, calendar integration, company-answer scoring, selection email, face-only F7).

2.2 Technical platform

Area	Technology
Operating system	Any system that runs Docker (developed and tested on Windows 11 with Docker Desktop)
Backend	Python 3.13, FastAPI, Pydantic v2, SQLAlchemy 2, Alembic migrations, Uvicorn
AI workflows	LangGraph with a PostgreSQL checkpointer; LangChain model interfaces
AI models	Google Gemini (gemini-3.5-flash, gemini-3.5-flash-lite, gemini-3.1-flash-lite), temperature 0; local Ollama 0.35.1 with qwen3-embedding:0.6b (embeddings) and qwen3:4b-instruct (screening fallback)
Database	PostgreSQL 17 with pgvector 0.8.7 (tables and vector search in one database)
Video	LiveKit server 1.13.9 (WebRTC), LiveKit client components in the browser
Speech-to-text	Deepgram Nova-3 streaming API (English)
Document and image processing	pypdf, python-docx, Tesseract OCR, OpenCV YuNet and SFace face models
Integrations	Three MCP servers (Python) for Blogger, Gmail (SMTP over TLS) and Google Calendar; Google Drive and Sheets APIs for the CV Collector
Frontend	React 19, Vite, Mantine 9, React Router, served by nginx

Area	Technology
Packaging	Docker Compose: 10 runtime services and 4 test services, named volumes for data
Quality tools	pytest (backend and MCP servers), Vitest (frontend), Ruff (lint and format)
Source control	Git on the BJIT GitHub repository, one branch per WBS module, BJIT commit message guideline

3. Functional Requirements

3.1 Overview

Figure 2 shows who uses CandiVue AI and for what. The interviewer and the candidate are on the left; the outside services the system works with are on the right. A dashed arrow marks a use case that is always part of another («include») or that can be added to it («extend»).

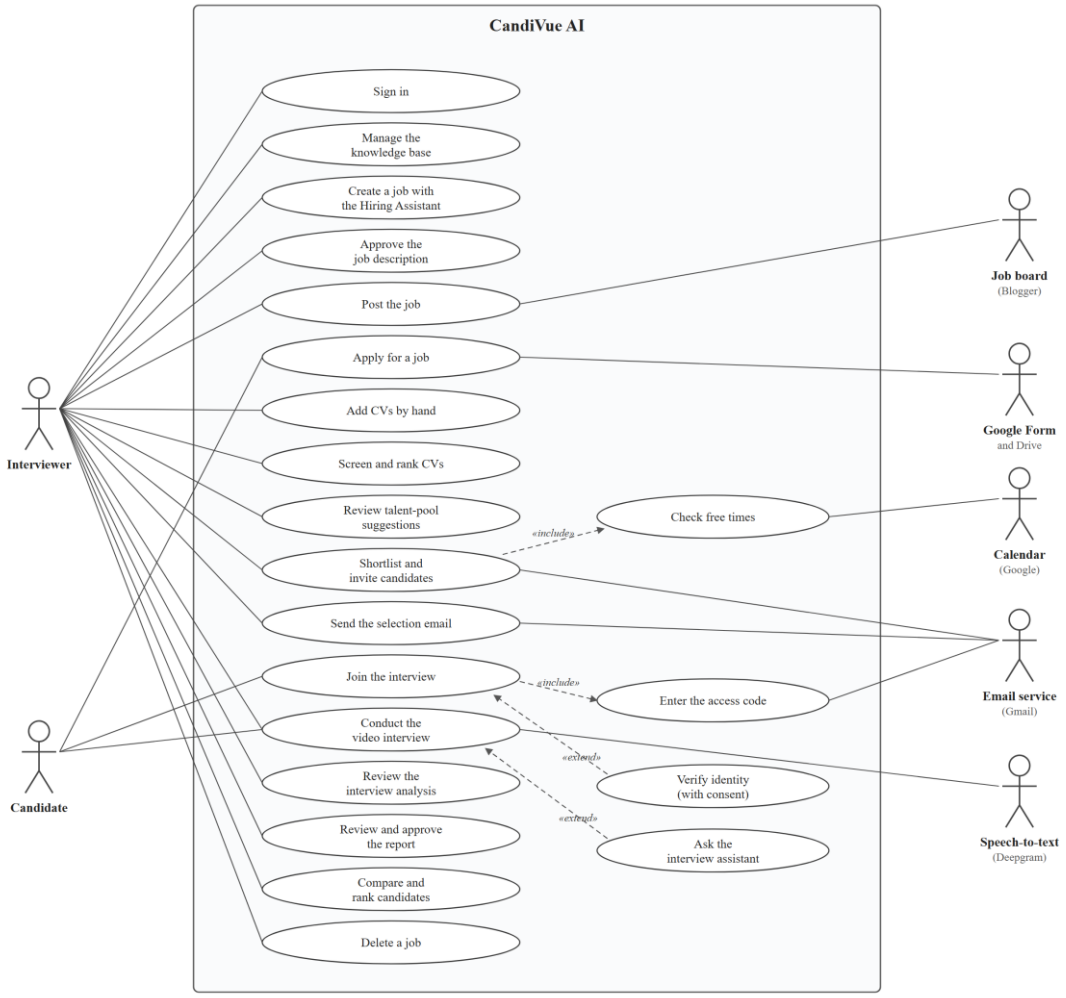


Figure 2: Use Case Diagram

The table below lists the main features of CandiVue AI. Each feature is described in the subsections that follow, together with its actors and requirements. Requirements are written with “shall” and numbered CVAI_FR_001 onwards; each one refers to the test cases that verify it (Appendix A).

Serial No	Main Features	Description
1	User access and authentication	Sign-in for interviewers and link-plus-code access for candidates
2	Knowledge base	Reference documents used by the AI (RAG)
3	Hiring Assistant: job profile and job description	AI chat that builds the job profile and writes the JD for approval
4	Job posting and CV collection	Publish the job; collect CVs from the form or by hand
5	CV screening and ranking (F1)	Batch reading, requirement matching, hard-requirement gate and ranking
6	AI-based candidate scoring (F2)	0–100 score on four weighted dimensions with justifications

Serial No	Main Features	Description
7	Talent rediscovery (F11, innovation)	Suggest consenting past applicants for a new vacancy
8	Shortlisting and interview scheduling (F3)	Shortlist in the chat, plan times, send invitations, manage the calendar
9	Candidate identification and profile overview (F7)	Check-in with camera pictures compared to the application photo
10	Live video interview (F3)	One-to-one browser video call with waiting room and chat
11	Real-time speech recognition and transcription (F4)	Live, speaker-attributed, timestamped transcript
12	Interview room assistant	Question suggestions for the interviewer during the call
13	Interview Q&A analysis (F5, F8, F9, F10)	Answer evaluation, requirement coverage, CV claim checks, interview score and quality
14	Consolidated candidate assessment (F6)	Evidence-backed report with interviewer review, export and comparison
15	Final ranking and selection email	Rank approved candidates and notify the selected ones
16	Data management and dashboard	Deletion, retention clean-up and the home dashboard

3.1.1. User access and authentication

Interviewers sign in with an email address and password. Candidates have no account: they open the private link from their invitation email and confirm it with a one-time code sent to the address they applied with. Every request carries a signed token, and the server checks the role on every route.

Actors: Interviewer, Candidate

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_001	The system shall let an interviewer sign in with email and password and issue a signed access token that expires after 8 hours.	Correct credentials open the app. A wrong email and a wrong password give the same message. Email case is ignored.	Only the Interviewer role has accounts. No self-registration or password reset.	TC_AUTH_01, TC_AUTH_02
CVAI_FR_002	The system shall create the first interviewer account from the configuration when the database has no user.	A fresh installation can be signed into without manual database steps. No second account is created on restart. Passwords shorter than 10 characters are refused.	Further interviewer accounts are not managed in the UI.	TC_AUTH_03
CVAI_FR_003	The system shall open a candidate's interview only with the private link and a one-time 6-digit code emailed to the applicant's address.	The code works once, for 10 minutes, with at most 5 wrong tries. A new code can be requested after 60 seconds. A forwarded link without the code gives no access.	Requires working email delivery. The address shown on screen is partly hidden.	TC_AUTH_04

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_004	The system shall refuse unknown interview links and links of interviews that have ended or were cancelled.	An unknown link shows “We don’t recognise this link”. A finished interview shows “The interview has ended”.	None.	TC_AUTH_05
CVAI_FR_005	The system shall enforce access rights on the server for every API route, so that candidates can never read scores, rankings, transcripts, analyses or reports.	A candidate token receives HTTP 403 on every interviewer route. An automated test checks every route of the running application.	Hiding a screen in the browser is not counted as protection.	TC_AUTH_06
CVAI_FR_006	The system shall sign the user out when the token has expired and explain why.	An expired session returns the interviewer to the sign-in page with a short explanation. Each browser tab keeps its own session.	Closing the tab also signs out.	TC_AUTH_07

3.1.2. Knowledge base

The interviewer uploads the organisation’s reference material into five collections: Reference library, Company policies, Previous job profiles, Email templates and Interview question bank. The text is split into passages and indexed with a local embedding model so the AI can find relevant passages by meaning.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_007	The system shall accept PDF, DOCX, TXT and Markdown documents up to 20 MB into a chosen collection and decide the file type from the file content.	Other types, renamed files and files without text are refused with a plain reason. The upload answers at once and the document shows “Indexing”.	Scanned PDFs without a text layer are not read here. The original file is not stored.	TC_KB_01, TC_KB_02
CVAI_FR_008	The system shall split each document into overlapping passages and index them with a local embedding model in the background.	Passages are about 1000 characters with a 150-character overlap. The document shows “Ready” when all passages are indexed, or a failure message with “Try again”, which continues where it stopped.	Indexing speed depends on the host CPU.	TC_KB_02, TC_KB_03
CVAI_FR_009	The system shall find the passages closest in meaning to a question, optionally within given collections, and return each with its source file and a score from 0 to 1.	Only fully indexed documents are searched. The “Try a search” box on the page shows the same results the AI would receive.	Results are ranked by meaning, not by keyword.	TC_KB_04
CVAI_FR_010	The system shall let the interviewer remove a document together with all its passages.	After removal the AI can no longer find any of its text. Candidates cannot use the knowledge base.	None.	TC_KB_05

3.1.3. Hiring Assistant: job profile and job description

Each job has its own chat thread with the Hiring Assistant (LangGraph 1). The interviewer describes the vacancy in plain words; the assistant saves a structured job profile, writes the job description using the knowledge base where useful, and waits for the interviewer's approval before anything is published. The same chat later handles posting, deadlines, shortlisting and invitations.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_011	The system shall create a structured job profile from the interviewer's description: requirements marked mandatory, required or preferred, minimum years of experience and suggested scoring weights.	The profile appears as a card in the chat and on the job page. Weights the interviewer states are kept; otherwise they are a suggestion that can be changed.	Multiple jobs are supported, each with its own thread.	TC_HA_01
CVAI_FR_012	The system shall draft the job description and, when useful, retrieve passages from the reference library, company policies and previous job profiles, showing the documents it used.	The draft shows "Based on" chips naming the source documents. If no passage matches in the chosen collection, all collections are searched.	The assistant decides when to search; it is not forced to.	TC_HA_02
CVAI_FR_013	The system shall pause until the interviewer approves, edits or rejects each job description draft.	Approve saves the version. Reject requires feedback and produces a rewritten version. An edit creates a new version that still needs approval. The chat input is locked while a decision is pending.	Only an approved description can be posted.	TC_HA_03
CVAI_FR_014	The system shall record the application deadline in local time and update the posted job when it changes.	A time in the past is refused. The new date appears in the job post.	Time zone is set by APP_TIMEZONE (Asia/Dhaka by default).	TC_HA_04
CVAI_FR_015	The system shall answer only hiring-related requests and reply to off-topic messages with a fixed message, without answering them.	Questions such as general knowledge, jokes or "ignore your instructions" get the fixed reply. A short answer to the assistant's own question is accepted. A mixed message is handled for its job part only.	The decision is made by the model through a dedicated tool; no extra AI request is spent on off-topic messages.	TC_HA_05
CVAI_FR_016	The system shall let the interviewer choose the AI model from an allowed list and show today's usage and limits.	Each model shows requests today and, once known, its daily limit and when it can be used again. A model outside the list is refused.	The provider does not report remaining quota, so the system counts its own requests.	TC_HA_06

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_017	The system shall keep the conversation state between sessions and report AI failures in plain words without losing the conversation.	An AI outage returns a clear message (HTTP 503). “Try again” continues the saved turn. Very long messages are refused. A turn that never ends is stopped with a message.	None.	TC_HA_07

3.1.4. Job posting and CV collection

After approval and the interviewer’s confirmation, the job is posted to a private Blogger blog through the job-board MCP server. The post links to a Google Form with the job already filled in. The CV Collector reads new form answers, downloads the CV files from Google Drive with read-only access and adds them to the job. CVs received another way can be added by hand.

Actors: Interviewer, Applicant (via the form)

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_018	The system shall post an approved job description to the job board through MCP, including the application form link, after the interviewer confirms.	The post shows the description, the closing date and an Apply link that pre-fills the job. A failed post keeps the description and can be retried; each call is tried up to three times.	Requires a Google Form link and Blogger credentials. Posting to commercial job boards is out of scope.	TC_COL_01
CVAI_FR_019	The system shall collect each new CV from the form answers once, on a configurable interval (default 5 minutes), and link it to the right job.	Answers without a job reference, consent or CV are skipped. A file already collected is never downloaded again. One failing answer does not stop the others.	Applicants must sign in with a Google account to upload a file in Google Forms.	TC_COL_02
CVAI_FR_020	The system shall accept only PDF and DOCX CVs up to 10 MB, decided by file content, and record the reason for every refused file.	A wrong type or an oversized file is refused before saving; the size is checked before downloading. Refused files appear as “File refused” with the reason.	None.	TC_COL_03
CVAI_FR_021	The system shall not collect or accept CVs sent after the application deadline.	A CV sent in time is collected even if it is processed after the close. Manual upload is refused after the deadline.	The interviewer can reopen applications by changing the deadline.	TC_COL_04
CVAI_FR_022	The system shall let the interviewer add CVs by hand with the same checks, one file per request, and record whether the applicants agreed to the talent pool.	Each file shows “Added” or the reason it was refused. Hand-added CVs are marked “Added by hand”.	At most 30 files per request.	TC_COL_05

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_023	The system shall list a job's applicants and let the interviewer view a CV or remove an applicant.	PDFs open in the browser, Word files download. Removal asks for confirmation. Candidates cannot see or change any of this.	None.	TC_COL_06

3.1.5. CV screening and ranking (F1)

The interviewer starts a screening of all CVs of a job from the job page. LangGraph 2 runs in the background: it reads every CV, extracts the facts, matches them against the job requirements by meaning, excludes candidates who fail a mandatory requirement and ranks the rest. Rules and local models do most of the work; Gemini is asked only for the few cases the rules cannot settle.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_024	The system shall screen all CVs of a job as one background batch and show its progress.	The page shows "n of N CVs". Only one screening per job runs at a time. A job without a profile or CVs cannot be screened.	Long batches depend on the host CPU.	TC_SCR_01
CVAI_FR_025	The system shall extract text from PDF and DOCX CVs and fall back to OCR when a PDF has no text layer.	A scanned CV is read and marked "Scanned CV, read with text recognition". An unreadable CV is skipped with a reason and the batch continues.	OCR quality depends on the scan.	TC_SCR_02
CVAI_FR_026	The system shall extract contact details, education, work history with dates, skills, certifications and projects from each CV.	Years of experience are counted from job dates, with overlapping jobs counted once. Many date formats are read (e.g. "03/2022 – now", "Till Date").	English CVs only.	TC_SCR_03
CVAI_FR_027	The system shall mark each job requirement as met, partly met or unmet for every candidate, with the CV line that supports it, using semantic matching.	Related wording counts (e.g. "Laravel Sail" for Docker); denials and wishes do not ("no Docker yet", "want to learn").	Matching uses local embeddings and rules, not the LLM.	TC_SCR_04
CVAI_FR_028	The system shall exclude a candidate who fails a mandatory requirement or the minimum years of experience, and show the reason.	Excluded candidates are listed separately with the reason and do not proceed. Unknown years never exclude a candidate.	Follows the trainer clarification that hard-requirement failures are excluded.	TC_SCR_05

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_029	The system shall use an AI check only where the rules cannot decide: reading the years of a CV without dates, and a second look before excluding for a missing mandatory requirement. Gemini answers first and the local model is the fallback.	The CV text is sent without name, email, phone or links. The answer must quote a real CV line. Answers are saved, so repeat runs send nothing. If both models fail, the run stops with a clear message.	The cloud model is paused for 10 minutes after a failure.	TC_SCR_06
CVAI_FR_030	The system shall present the remaining candidates as an ordered ranking with their scores and pipeline stage.	On the synthetic test set the top, middle and excluded groups match the documented answer key.	Order within a group may change slightly between models.	TC_SCR_07

3.1.6. AI-based candidate scoring (F2)

Every candidate who passes the gate receives a score from 0 to 100 on four dimensions and an overall weighted score. The weights are suggested for each job and can be changed by the interviewer.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_031	The system shall score technical skills, relevant experience, qualifications and an ATS-style CV–JD match from 0 to 100, each with a short justification, and combine them into a weighted overall score.	The candidate panel shows each dimension, its weight, its reason and the overall score.	Exact formulas are documented in the README.	TC_SCO_01
CVAI_FR_032	The system shall use per-job weights suggested from the job profile, which the interviewer may change or reset.	Weights must add up to 100. Interviewer weights take priority. A change shows “The weights changed since the last screening” with “Screen again”.	None.	TC_SCO_02
CVAI_FR_033	The system shall give the same scores for the same CVs and job profile on every run.	Two runs on the same data give identical scores. AI temperature is 0 and saved AI answers are reused.	A different AI model could give different answers for the few AI-checked cases.	TC_SCO_03
CVAI_FR_034	The system shall store the scores so the assessment report can use them.	The report shows the CV screening score and dimensions of the same run.	None.	TC_SCO_04

3.1.7. Talent rediscovery (F11, innovation)

Applicants who agreed to be considered for other vacancies form a talent pool. After each screening, and on request, the pool is screened against the job with the same checks as LangGraph 2, and suitable

people are suggested on the job page. The interviewer decides whether to add a suggestion; it is never added automatically.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_035	The system shall include in the talent pool only applicants who agreed, applied within the last 365 days, are not in an active hiring step and have not already applied to this job.	Each person is checked once, using their latest application.	Consent comes from the form question or from the interviewer for hand-added CVs.	TC_TAL_01
CVAI_FR_036	The system shall screen the pool against the job after each screening and on “Check again”, and suggest applicants who pass the gate with a score of at least 50.	Suggestions show the job they applied for, the score and the reasons. Only one check runs at a time.	No extra AI request is made beyond the screening checks.	TC_TAL_02
CVAI_FR_037	The system shall add a suggestion to the job only when the interviewer chooses to, by copying the application.	The original application stays unchanged; the copy is marked “From talent pool”.	None.	TC_TAL_03
CVAI_FR_038	The system shall let consent be withdrawn at any time, after which the person is no longer suggested.	Removing someone from the pool hides their suggestions. Candidates cannot see the pool.	None.	TC_TAL_04

3.1.8. Shortlisting and interview scheduling (F3)

The interviewer shortlists candidates and plans interviews in the Hiring Assistant chat. The system checks every rule before anything is saved, shows an invitation card for confirmation, then creates each interview with a private link, sends the invitation email and adds the event to the interviewer’s Google Calendar.

Actors: Interviewer, Candidate (receives email)

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_039	The system shall build and change the shortlist from chat requests (top N, score criteria, add, remove), using ranked candidates only.	Ties at the cut are mentioned. Excluded or unknown names are refused with the reason.	None.	TC_SCH_01
CVAI_FR_040	The system shall allow invitations only after the application deadline has passed and a screening includes every applicant.	A request before that is refused with what to do next.	This ensures all applicants are compared before interviews start.	TC_SCH_02

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_041	The system shall check proposed times: at least 1 hour ahead, within 90 days, no overlap with other interviews and no busy time in the interviewer's Google Calendar; outside working hours gives a warning.	Free slots skip weekends and booked times. If the calendar cannot be reached, only CandiVue's own interviews are checked and the card says so.	Working hours are 09:00–17:00 local time.	TC_SCH_03
CVAI_FR_042	The system shall show an invitation card with the candidates, times and email preview, and send nothing until the interviewer confirms.	Cancel saves nothing. Problems found are returned to the assistant to fix. The checks run again on confirmation.	None.	TC_SCH_04
CVAI_FR_043	The system shall create each confirmed interview with a unique private link, email the invitation using the knowledge-base template and add the event to the calendar.	The email includes a calendar invite (.ics). Without a template a standard text is used. A failed email keeps the interview and can be resent from the Interviews page. All template text is HTML-escaped.	In development all email goes to a test recipient.	TC_SCH_05
CVAI_FR_044	The system shall reschedule and cancel interviews, notifying the candidate and updating the calendar.	A moved interview gets a new link and updates the same calendar event. A cancelled interview's link stops working and a cancellation email is sent.	None.	TC_SCH_06

3.1.9. Candidate identification and profile overview (F7)

Applicants may send a face photo with their application and give separate consent for its use. When they are invited, the photo is processed into an encrypted face reference. At check-in, after the access code, the candidate takes three camera pictures that are compared with the reference on the server. The interviewer sees the result before letting the candidate in; the candidate never does.

Actors: Candidate, Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_045	The system shall process an applicant's photo only after they are invited and only with their consent, keeping an encrypted face embedding and a small encrypted copy of the photo, never the original file.	The photo is checked by content (JPEG, PNG, WebP, HEIC), turned upright and must show exactly one clear face. An unusable photo keeps only the reason (no face, several faces, too small, unreadable, too large).	Requires FACE_REFERENCE_KEY. The original stays in the form owner's Google Drive.	TC_IDV_01
CVAI_FR_046	The system shall let the candidate check in with three camera pictures after visible consent, or skip the check without penalty.	Each picture must show one clear face; otherwise the candidate is asked to retake it. After too many tries the candidate continues without the check.	Declining deletes the photo data at once.	TC_IDV_02

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_047	The system shall compare the check-in pictures with the application photo locally and show the interviewer the photo, the result and a match score in percent.	Results: match, not confirmed, different person, looks like another applicant of the job, or why no comparison was possible. The candidate profile (name, headline, key skills, CV score, areas to probe) is shown only for a confident match.	The score is a fixed scale on face similarity, not a probability. It proves a match with the application photo, not legal identity.	TC_IDV_03
CVAI_FR_048	The system shall never show the identity result to the candidate and never block the interview because of it.	Candidate routes do not return the result. The interviewer decides whether to continue.	None.	TC_IDV_04
CVAI_FR_049	The system shall delete face data 30 days after the interview, on cancellation, on withdrawal and with the candidate or job.	Expired data is removed by the hourly clean-up. Recent interviews keep their data.	Retention is set by FACE_RETENTION_DAYS.	TC_IDV_05

3.1.10. Live video interview (F3)

Interviewer and candidate meet in a browser video call served by LiveKit. The candidate waits in a waiting room until the interviewer lets them in. The interviewer's screen also shows the CV, the live transcript, the identity result and the room assistant.

Actors: Interviewer, Candidate

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_050	The system shall give each participant a call ticket valid only for their own interview room.	A ticket names the person and their role. Candidates can see only their own interview. A missing video setup is reported clearly.	One interviewer and one candidate per interview.	TC_VID_01
CVAI_FR_051	The system shall let the call start only after the invitation was emailed, and keep the candidate in a waiting room until the interviewer lets them in.	Both sides see whether the other is present. Nobody is let in after the interview has ended.	None.	TC_VID_02
CVAI_FR_052	The system shall provide in-call controls: mute, camera on/off, screen sharing and, for the interviewer, end session.	Each change is visible to the other side within a few seconds. Ending asks for confirmation.	Device permissions are controlled by the browser.	TC_VID_03
CVAI_FR_053	The system shall reconnect a participant to the same session automatically after a dropped connection or page reload.	A reloaded candidate returns to the call without waiting again. The transcript captured so far is kept.	Long network outages end the call on the participant's side until the network returns.	TC_VID_04

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_054	The system shall record the interview's participants and start and end times, and close the call and the link when it ends.	The first join marks the interview live (from a signed LiveKit webhook; forged webhooks are refused). Ending disconnects everyone and starts the analysis.	None.	TC_VID_05
CVAI_FR_055	The system shall provide a text chat between interviewer and candidate during the call.	Messages are limited in length and rate and stop when the interview ends.	None.	TC_VID_06
CVAI_FR_056	The system shall show the interviewer a CV overview panel during the call and never show it to the candidate.	The panel shows the parsed CV and screening summary.	None.	TC_VID_07

3.1.11. Real-time speech recognition and transcription (F4)

A transcription worker joins each live call as a hidden listener. It streams each person's microphone separately to Deepgram, so the speaker is known from whose audio it is. Words appear on the interviewer's screen while they are spoken, and finished turns are saved with their times.

Actors: Interviewer (reader), Candidate (speaker)

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_057	The system shall transcribe both participants during the call and attribute each turn to the interviewer or the candidate from the participant's identity.	Each microphone track is one speaker. The listener cannot publish audio or video. Live text is sent to the interviewer only; candidates can never read the transcript.	English only.	TC_TRN_01
CVAI_FR_058	The system shall save the transcript as an ordered list of turns with start and end times measured from the start of the interview.	A pause of 1.2 seconds or more starts a new turn; very long answers are split at 45 seconds. A growing turn updates one row. Overlapping speech becomes separate turns.	Times follow when the audio was heard, also after mute or reconnect.	TC_TRN_02
CVAI_FR_059	The system shall not lose speech when the connection to the speech service drops.	Audio waits in memory (up to 120 seconds) and is sent first after reconnecting; nothing is repeated.	Audio older than 120 seconds of outage is dropped with a log warning.	TC_TRN_03
CVAI_FR_060	The system shall keep the transcript after the interview, including the last words spoken before the end.	The final words are saved when the call ends; if the service is too slow, the last live words are kept.	None.	TC_TRN_04
CVAI_FR_061	The system shall show the transcription status and any problem in plain words in the interview room.	States include listening, not set up, refused key and no credit. A refused key is not retried.	None.	TC_TRN_05

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_062	The system shall use the job's requirement names to improve recognition of technical terms and shall not store raw audio.	Audio is processed in memory only. Speech-service data use for training is switched off.	None.	TC_TRN_06

3.1.12. Interview room assistant

During the call the interviewer can ask the Hiring Assistant for questions from the organisation's question bank, for questions based on the candidate's CV, or for a short candidate summary. The candidate never sees the assistant.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_063	The system shall suggest interview questions from the question bank (RAG) and from the candidate's CV, and keep the CV-based questions for the interview.	Suggestions can be copied. CV questions are made once per interview and saved. Off-topic requests get the fixed reply.	Suggested questions do not change the analysis, which uses the questions actually asked.	TC_RAS_01

3.1.13. Interview Q&A analysis (F5, F8, F9, F10)

When the interview ends, LangGraph 3 runs in the background. It splits the transcript into question-and-answer units, checks answers against the knowledge base, evaluates them, maps them to the job requirements, compares CV claims with the answers, computes an interview score and reviews the quality of the interview itself.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_064	The system shall start the analysis automatically when an interview ends, show its progress and continue a failed analysis from where it stopped.	Only ended interviews are analysed. "Analyse again" replaces the analysis and its report.	About three AI requests per typical interview.	TC_QA_01
CVAI_FR_065	The system shall split the transcript into Q&A units, separate multi-part questions (e.g. Q3a, Q3b) and link follow-ups to their question.	Question lines must be the interviewer's and answer lines the candidate's. Greetings and small talk stay in the transcript but are not scored.	At most 60 units per interview.	TC_QA_02

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_066	The system shall evaluate each job- or company-related answer for relevance, factual accuracy, completeness and articulation (0–100), each with a justification and the transcript lines it rests on.	Answers are evaluated in batches of up to 20 per request. Lines that do not belong to the unit are dropped.	Accent, grammar and filler words are not penalised.	TC_QA_03
CVAI_FR_067	The system shall check answer accuracy against the knowledge base: job answers against the Reference library and company answers against Company policies.	Only passages scoring at least 0.5 are used; the answer is labelled with the document it was checked against, or “not checked against a reference document”.	Scoring company answers is a project decision awaiting trainer confirmation.	TC_QA_04
CVAI_FR_068	The system shall map the units to the job requirements and list the requirements that were never explored.	Coverage is computed in code from the evaluations, so it is exact.	None.	TC_QA_05
CVAI_FR_069	The system shall compare important CV claims with the interview and mark each as supported, contradicted, unverified or never discussed, with a neutral follow-up question (F8).	Every claim points to a real CV item and, except “never discussed”, to the candidate’s own lines. Flags are prompts for a person, never accusations.	None.	TC_QA_06
CVAI_FR_070	The system shall compute an interview score separate from the CV score on six dimensions with evidence (F9).	Technical competency, communication, response relevance, problem-solving, experience depth and job alignment; a dimension without evidence says so.	None.	TC_QA_07
CVAI_FR_071	The system shall assess the quality of the interview (good, needs attention, poor) with reasons and suggestions, and compare it with the job’s other interviews (F10).	Reasons include the interviewer’s share of talking time, requirement coverage and questions that gave little evidence. A poor interview offers rescheduling through the Hiring Assistant.	Rescheduling stays the interviewer’s decision.	TC_QA_08
CVAI_FR_072	The system shall remove names, email addresses, phone numbers and links from transcripts and CV facts before they are sent to the AI provider.	Prompts contain numbered lines without personal identifiers.	None.	TC_QA_09

3.1.14. Consolidated candidate assessment (F6)

After the analysis, LangGraph 4 prepares the assessment report. It combines the CV screening, the scores and the interview analysis into one position, writes the conclusions with their evidence, and pauses for the interviewer’s review. The approved report can be exported and compared with other candidates.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_073	The system shall prepare the report automatically after the analysis, merging CV screening, candidate scores, the Q&A analysis, CV claims, interview score, interview quality and the identity result.	The report page shows the steps while it is prepared and a notification when it is ready.	A report needs a finished analysis.	TC_REP_01
CVAI_FR_074	The system shall present an overall suitability position backed by component scores, not a single opaque number.	Strong fit from 80, good fit from 65, partial fit from 50, otherwise not yet suitable. Without scored answers the report states that it rests on the CV alone.	The position is advice; the interviewer decides.	TC_REP_02
CVAI_FR_075	The system shall list strengths, weaknesses, evidence gaps and areas to verify, each citing CV items, timestamped transcript lines or requirement records.	Citations that do not exist are removed. A conclusion without evidence is repaired up to twice and then marked as unsupported, never silently kept.	At most four conclusions per list.	TC_REP_03
CVAI_FR_076	The system shall pause the report for the interviewer to approve it or request changes with a comment, and write the next version from that comment.	A change request needs a comment. The new version states what changed. The reviewer and approval time are recorded.	None.	TC_REP_04
CVAI_FR_077	The system shall export the report as a printable, self-contained HTML file that can be saved as PDF.	Sources are numbered references. A draft is marked “Not yet approved” on every printed page; an approved report names the approver.	None.	TC_REP_05
CVAI_FR_078	The system shall show assessed candidates of the same job side by side.	The comparison lists the position, component scores and dimensions for each chosen candidate.	None.	TC_REP_06
CVAI_FR_079	The system shall show the source of any conclusion on request and keep reports from candidates.	Clicking a source chip shows the CV text or transcript line. Candidate tokens receive HTTP 403 on all analysis and report routes.	None.	TC_REP_07

3.1.15. Final ranking and selection email

The Reports page ranks the candidates whose reports are approved and lets the interviewer send a selection email to the candidates they choose. The email is a notification only; offer details are handled outside CandiVue.

Actors: Interviewer, Candidate (receives email)

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_080	The system shall rank candidates with an approved report by overall score, then interview score, with equal candidates sharing a rank.	Candidates without an approved report are listed with the reason. A candidate interviewed twice counts once.	None.	TC_SEL_01
CVAI_FR_081	The system shall send a selection email only after the report is approved, from the knowledge-base template or a standard text, after the interviewer has checked and possibly edited it.	The preview shows the email as the candidate will see it. It is sent once; the candidate is then marked “Selected”. A failed email does not mark the candidate.	BID §4.2 excludes offer management; this notification needs trainer approval.	TC_SEL_02

3.1.16. Data management and dashboard

Interviewers can delete an applicant or a whole job with everything that belongs to it. A background clean-up removes data whose retention has ended. The home dashboard shows what needs attention.

Actors: Interviewer

Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_FR_082	The system shall delete an applicant and all associated data: scores, passages, interviews, transcripts, analyses, reports, face data, the CV file and saved workflow state.	One request removes everything; related rows are deleted by database cascades.	Data already sent to an external provider cannot be recalled.	TC_DM_01
CVAI_FR_083	The system shall delete a job after confirmation, cancelling its scheduled interviews, removing the job post and deleting all its data.	Deletion is refused while a screening, rediscovery, analysis or report is running or an interview is live. A job-board or calendar failure is reported but does not stop the deletion.	None.	TC_DM_02
CVAI_FR_084	The system shall show a home dashboard with counts, the next steps, upcoming interviews and applications per day.	Live interviews appear first in the next steps. Counts and days use local time.	Read-only; actions stay on their own pages.	TC_DM_03
CVAI_FR_085	The system shall remove expired face data and workflow state that is no longer needed, every hour.	Finished runs delete their saved state at once; the hourly job removes the rest.	None.	TC_DM_04

4. User Interface

The application has one interface for interviewers, protected by sign-in, and one page for candidates reached through the private interview link. The screens are listed below; the visual design follows the design handoff in docs/design_handoff_candivue_ui.

UI No.	UI Name	Related Function Req ID	Description	Test case Identifier
CVAI_UI_001	Sign-in page	CVAI_FR_001 to 006	Email and password form with the product statement. Shows why the user was signed out after an expired session.	TC_AUTH_01, TC_AUTH_07
CVAI_UI_002	Home dashboard	CVAI_FR_082 to 085	Counts, next steps (live interviews first), upcoming interviews and a 14-day applications chart.	TC_DM_03
CVAI_UI_003	Jobs list	CVAI_FR_011 to 017	All jobs with CVs received, screening state, interviews and last change; opens a job or a new assistant chat.	TC_HA_01
CVAI_UI_004	Hiring Assistant	CVAI_FR_011 to 017, CVAI_FR_039 to 044	Chat per job with profile, JD review, posting, deadline, shortlist and invitation cards; model picker with daily usage.	TC_HA_01 to TC_HA_07, TC_SCH_01 to TC_SCH_06
CVAI_UI_005	Job page	CVAI_FR_018 to 023, CVAI_FR_024 to 030, CVAI_FR_031 to 034, CVAI_FR_035 to 038	Applicants, Add CV, scoring weights, screening progress, ranking with candidate panel, excluded list, talent-pool suggestions and Delete job.	TC_COL_05, TC_SCR_01, TC_SCO_02, TC_TAL_02
CVAI_UI_006	Knowledge base	CVAI_FR_007 to 010	Five collections, upload with progress and status, remove, retry and a “Try a search” box.	TC_KB_01 to TC_KB_05
CVAI_UI_007	Interviews list	CVAI_FR_039 to 044	Upcoming and past interviews with invitation state; join, resend and cancel; links to analysis and report.	TC_SCH_05, TC_SCH_06
CVAI_UI_008	Interview room (interviewer)	CVAI_FR_050 to 056, CVAI_FR_057 to 062, CVAI_FR_063, CVAI_FR_045 to 049	Video call, waiting-room banner with the check-in result, CV panel, live transcript, chat and room assistant.	TC_VID_01 to TC_VID_07, TC_TRN_01, TC_RAS_01
CVAI_UI_009	Candidate interview page	CVAI_FR_001 to 006, CVAI_FR_045 to 049, CVAI_FR_050 to 056	Access code step, identity check-in, device check, waiting room and call. Shows no scores or results.	TC_AUTH_04, TC_IDV_02, TC_VID_02
CVAI_UI_010	Interview analysis	CVAI_FR_064 to 072	Q&A units with scores and sources, requirement coverage, transcript, CV claims, interview score, interview quality and session details.	TC_QA_01 to TC_QA_08
CVAI_UI_011	Assessment report	CVAI_FR_073 to 079	Position, component scores, conclusions with source chips and a source panel, review bar, export and comparison tab.	TC_REP_01 to TC_REP_07
CVAI_UI_012	Reports	CVAI_FR_080 to 081	Final ranking of approved candidates with comparison and selection email, then every report of the job.	TC_SEL_01, TC_SEL_02

5. Non-Functional Requirements

The general non-functional requirements are listed below. Figures marked as measured come from the verification on the development laptop (Windows 11, Docker Desktop, 16 CPU threads, 7.6 GB for Docker) with synthetic data, recorded in docs/VERIFICATION.md.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_NFR_001	Screen a batch of 20 CVs (parse, gate, score, rank) within 5 minutes on the reference laptop without a GPU.	Measured 3–5 minutes for 20–22 synthetic CVs on the first run; a repeat run takes 7–8 seconds because parsed CVs and embeddings are reused. The figure is recorded in the README.	The trainer gave no numeric limit; 5 minutes is the project target.	TC_NFR_01
CVAI_NFR_002	Show live transcript text with minimal delay.	Words appear about 1 second after they are spoken; a finished turn is saved within a median of 0.8 seconds (95% within 2.7 seconds).	Depends on the network to the speech service.	TC_NFR_02
CVAI_NFR_003	Answer non-AI API requests promptly, also while a screening runs.	Median below 100 ms and 95% below 200 ms during a 29-CV screening (measured 62 ms and 130 ms).	Measured on the development laptop.	TC_NFR_03
CVAI_NFR_004	Run long AI work in the background with visible progress.	Screening, indexing, rediscovery, analysis and report requests return at once; the page shows progress and a plain status.	Two background workers.	TC_NFR_04
CVAI_NFR_005	Support one live interview and a background CV screening at the same time.	A 45-minute interview stayed connected while 29 CVs were screened; 218 turns were saved in order; memory stayed flat.	One concurrent interview is the BID minimum.	TC_NFR_05
CVAI_NFR_006	Have the assessment ready soon after the interview ends.	The report of a 45-minute interview was ready 82 seconds after “End” using 6 AI requests.	Depends on the AI provider’s response time and quota.	TC_NFR_06
CVAI_NFR_007	Keep AI usage within the provider’s free tier.	About 4 AI requests per interview for analysis and report; screening asks the AI only for CVs the rules cannot settle, once.	Daily limits differ per model.	TC_NFR_07

5.1 Performance Requirements

Response time. Non-AI requests (lists, pages, status checks) must feel immediate: the target is a median below 100 ms and 95% below 200 ms on the reference machine, also while a screening is running. AI-backed chat turns may take several seconds; the interface shows that the assistant is working and allows up to 120 seconds.

Throughput. One screening batch of 20 CVs should finish within 5 minutes on a laptop CPU. Screening processes at most 4 CVs in parallel so that the database connection pool and the local model are not overloaded.

Capacity. The system is sized for one organisation, a handful of open jobs, a few hundred applicants and one live interview at a time, as stated in the BID. Two background workers allow a screening and an interview analysis to run together.

Degradation modes. If Gemini is unavailable or over its limit, screening continues with the local model; chat and analysis report a clear error and can be retried later without losing work. If the calendar cannot be reached, interviews are still scheduled with a warning. If email fails, the interview is kept and the invitation can be resent. If speech-to-text fails, the call continues and the room shows the problem.

Resource use. The reference set-up uses about 6–8 GB of memory in Docker, mostly for the local models. Raw audio and video are never written to disk.

5.2 Safety Requirements

CandiVue AI influences decisions about people, so its safety requirements protect candidates from unfair or unchecked automated outcomes and protect data from accidental loss.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_SAF_001	The system shall support the hiring decision and never take it: no candidate is rejected, invited, selected or emailed without an interviewer action.	Invitations, JD publication, report approval and the selection email each require an explicit confirmation.	The hard-requirement gate excludes candidates from the ranking, as required by the trainer; the interviewer sees the reason and can change the profile.	TC_SAF_01
CVAI_SAF_002	AI conclusions shall be traceable to evidence and labelled when unsupported.	Every score has a reason; every report conclusion cites CV or transcript evidence or is marked unsupported.	AI output can still be wrong; the interface tells the user to check sources.	TC_REP_03
CVAI_SAF_003	CV claim flags and identity results shall be presented as prompts for a person, never as accusations or automatic decisions.	Wording is neutral; the identity result never blocks the interview.	None.	TC_QA_06, TC_IDV_04
CVAI_SAF_004	A failure in one item shall not stop or corrupt the others.	One unreadable CV, failed download or failed email is reported and the rest continue.	None.	TC_SCR_02, TC_COL_02
CVAI_SAF_005	Destructive actions shall ask for confirmation and refuse to run while dependent work is in progress.	Removing an applicant, a document or a job shows a confirmation; job deletion is refused while runs are active.	None.	TC_DM_02
CVAI_SAF_006	Development and test runs shall never email real people.	With EMAIL_TEST_RECIPIENT set, every email goes to the test address with the intended recipient in the subject.	Must be removed for real use.	TC_SAF_02

5.3 Security Requirements

The system handles CVs, interview transcripts and face data, so it is designed to protect them against accidental and malicious access. The main measures are:

- Cryptography: Argon2 password hashes, HS256-signed tokens, SHA-256 hashes of interview links, HMAC-SHA256 access codes compared in constant time, and Fernet encryption of face data.
- Separation: each external integration runs in its own container with only its own credentials; the backend never sees them. The database is not exposed outside Docker.
- Restricted communication: the browser reaches only nginx; the API, database, models and MCP servers are reachable only inside Docker's network.
- Logging: logs contain ids and counts only, never names, emails, CV or transcript text; interview link tokens are masked in the web server log.
- Integrity: uploads are checked by content and size; downloaded face models are verified by SHA-256 checksums at build time.

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_SEC_001	Store passwords only as Argon2 hashes and give the same response time and message for unknown emails and wrong passwords.	No plain password in the database; timing does not reveal registered emails.	None.	TC_SEC_01
CVAI_SEC_002	Issue signed, expiring access tokens and accept only valid tokens with the required claims.	HS256, secret of at least 32 characters from the environment, 8-hour expiry; expired, forged or malformed tokens get HTTP 401.	Tokens are kept in the tab's session storage.	TC_SEC_02
CVAI_SEC_003	Enforce the Interviewer and Candidate roles on the server for every route.	Automated test of every route; candidates get HTTP 403 on all interviewer data.	None.	TC_AUTH_06
CVAI_SEC_004	Store interview links and access codes only as hashes.	A copy of the database does not reveal working links or codes.	None.	TC_SEC_03
CVAI_SEC_005	Validate all API input and every uploaded file.	Pydantic models with length limits on every text field; file type by content; size limits (CV 10 MB, document 20 MB, photo limits, 25 MB per request at nginx).	None.	TC_SEC_04
CVAI_SEC_006	Keep all secrets in environment variables and out of the repository.	.env is git-ignored; .env.example holds placeholders only; repository scan finds no keys.	None.	TC_SEC_05
CVAI_SEC_007	Never show stack traces, file paths, configuration or prompt text to the user.	Unexpected errors return "Internal server error"; AI errors return a plain message with HTTP 503.	Details are logged on the server only.	TC_SEC_06

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_SEC_008	Protect the browser session with security headers.	Content-Security-Policy, X-Frame-Options DENY, nosniff, no-referrer and a Permissions-Policy that allows camera and microphone only for CandiVue.	Plain HTTP on localhost is a documented development exception; HTTPS is enabled when a certificate is supplied.	TC_SEC_07
CVAI_SEC_009	Treat CVs, transcripts and documents as data, not instructions, in every AI prompt.	Prompts state this explicitly; the Hiring Assistant refuses out-of-scope requests.	Prompt injection cannot be ruled out completely; all AI actions that change data need confirmation.	TC_HA_05
CVAI_SEC_010	Pin all dependency and image versions and run containers without root rights.	Exact versions in requirements and package lock files; images pinned by version; application containers run as a non-root user.	None.	TC_SEC_08

6. Design Constraints

The following decisions are mandated by the BID, the trainer clarifications or the project set-up and must be followed:

- Language and framework: all backend and AI logic in Python 3.13 with FastAPI and Pydantic, with OpenAPI documentation generated automatically (BID §8.1).
- AI workflows: multi-step LangGraph graphs with explicit state, nodes and edges. LangGraph 1 is a tool-calling agent; the other graphs are fixed workflows so scores are repeatable. Workflow state is checkpointed in PostgreSQL and resumes after a restart.
- Deployment: the complete system runs in Docker containers defined in one Docker Compose file and starts with one documented command on a clean machine. Cloud hosting is not used.
- Data storage: PostgreSQL with the pgvector extension for both relational data and embeddings; named volumes for the database, CV files and local models.
- Integrations: the job board, email and calendar are reached through MCP servers; the CV Collector uses the Google Drive and Sheets APIs with a read-only service account.
- AI provider: Google Gemini on the free tier, so the design minimises requests; the provider and model are configuration settings, not code.
- Data: only the trainer’s sample CVs or synthetic data are used during development and the demonstration.
- Interviews: one interviewer and one candidate per interview; English only.
- Source control: the BJIT GitHub repository, one branch per WBS module, commits following the BJIT commit message guideline with Redmine references.
- Timeline and team: one developer, 20 working days, delivery by 27 October 2026.

7. Software Quality Attributes

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_SQA_001	**Explainability.** Every score, rank, flag and conclusion shows its reason and the CV or transcript evidence behind it.	No unexplained number appears on any screen; report citations open their source.	AI explanations are generated and may be imperfect.	TC_REP_03, TC_SCO_01
CVAI_SQA_002	**Reproducibility.** The same CVs and job profile give the same screening result.	Repeated runs give identical scores (temperature 0, rules, saved AI answers).	Changing the AI model may change AI-checked cases.	TC_SCO_03
CVAI_SQA_003	**Reliability and recovery.** Long-running work survives failures and restarts.	Interrupted runs resume at start-up from the last completed step; finished work is not repeated.	None.	TC_SQA_01
CVAI_SQA_004	**Maintainability.** Clear separation of concerns and readable code.	One module per provider; plain-language docstrings; no dead code blocks; Ruff lint clean.	None.	TC_SQA_02
CVAI_SQA_005	**Testability.** Core business logic is covered by automated tests that run in Docker against the real database.	371 backend, 27 MCP server and 53 frontend tests pass.	External services are replaced by fakes in tests.	TC_SQA_03
CVAI_SQA_006	**Usability.** Screens are understandable without explanation and errors are written for non-technical users.	Progress is shown for long tasks; every error message says what happened and what to do.	None.	TC_SQA_04
CVAI_SQA_007	**Portability.** The system starts on any machine that runs Docker.	A clean clone starts with the documented commands only.	Requires about 8 GB of memory for Docker.	TC_SQA_05
CVAI_SQA_008	**Fairness.** Scoring uses job-related criteria only.	No column for age, gender, ethnicity, nationality, religion or marital status; prompts forbid using or inferring them, accent or grammar.	Model bias cannot be fully excluded; limits are documented in the README.	TC_SQA_06

8. User Interface

This section states the rules that apply to every screen listed in section 4.

UI No.	UI Name	Related Info No.	Description	Notes	Test case Identifier
CVAI_UI_101	Responsive layout	Design handoff, all screens	Every screen works on phone, tablet and desktop. Below 992 px the menu becomes a drawer; tables become stacked cards.	Overrides the handoff's desktop-first note.	TC_UI_01
CVAI_UI_102	Light and dark theme	Theme tokens	The user can switch between light and dark themes; the choice is remembered on the device.	None.	TC_UI_01

UI No.	UI Name	Related Info No.	Description	Notes	Test case Identifier
CVAI_UI_103	Progress and AI activity	AiSignalStatus component	Long tasks show a progress bar or the current step and elapsed time while the AI works.	None.	TC_SQA_04
CVAI_UI_104	Plain-language messages	All screens	Errors explain what happened and what to do next; success is confirmed with a short notification.	No technical details are shown.	TC_SEC_06
CVAI_UI_105	Confirmations	Remove and delete actions	Actions that cannot be undone ask for confirmation and say what will be removed.	None.	TC_DM_02
CVAI_UI_106	Accessibility practice	All screens	Labelled controls, keyboard use, visible focus and sufficient contrast.	Formal WCAG certification is out of scope.	TC_UI_01

9. Other Requirements

REQUIREMENT ID	Requirement Description	Acceptability/ Completion Criteria	Limitations/ Constraints	Test case Identifier
CVAI_OR_001	Privacy disclosure: the README shall list each external provider that receives candidate data and exactly what it receives.	Gemini, Deepgram, Gmail, Google Calendar, Google Forms/Drive/Sheets and Blogger are listed with the data sent.	None.	TC_OR_01
CVAI_OR_002	Data minimisation and retention: store only what the features need and delete it when no longer needed.	No raw audio or video; only extracted document text; face data deleted after 30 days; workflow state deleted after use.	Data in the form owner's Google Drive is outside CandiVue's control.	TC_DM_04, TC_IDV_05
CVAI_OR_003	Documentation: the repository shall contain a README with purpose, architecture, setup, configuration, usage and known limitations, and the API documentation shall be available through OpenAPI.	README complete; /docs on the backend lists every route.	None.	TC_OR_02
CVAI_OR_004	Delivery: the final evaluated commit shall be tagged and the documentation deliverables committed under /docs.	Tag v1.0-final on the default branch; SRS, WBS, innovation rationale and verification report in /docs.	None.	TC_OR_03
CVAI_OR_005	Repository hygiene: no secrets, personal candidate data, virtual environments or build output in the repository.	.gitignore covers .env, CV intake data and build folders; only synthetic test data is committed.	None.	TC_SEC_05

Appendix A Test Case Index

Each test case identifier used in this document is listed below with the tests that verify it. Automated tests are in backend/tests (pytest), the MCP server folders and frontend/src (Vitest); they run with “docker compose run --rm tests”, the MCP test services and “npm test”. Manual checks and measurements are recorded in docs/VERIFICATION.md.

Test case ID	Verified by	Type
TC_AUTH_01	test_auth_api: test_interviewer_signs_in, test_email_is_not_case_sensitive, test_failed_sign_in_gives_the_same_message	Automated
TC_AUTH_02	test_security: test_token_round_trip, test_expired_token_is_rejected, test_token_signed_with_another_secret_is_rejected	Automated
TC_AUTH_03	test_auth_api: test_first_interviewer_is_created_from_settings, test_no_second_account_is_created	Automated
TC_AUTH_04	test_auth_api: test_candidate_joins_with_the_link_and_the_emailed_code, test_a_forwarded_link_without_the_code_gets_nowhere, test_an_expired_code_is_replaced_by_a_new_one	Automated
TC_AUTH_05	test_auth_api: test_unknown_link_is_refused, test_link_stops_working_after_the_interview	Automated
TC_AUTH_06	test_auth_routes: test_every_api_route_is_guarded, test_candidate_cannot_see_scores, test_candidate_route_rejects_interviewer	Automated
TC_AUTH_07	client.test.js: an expired session signs the interviewer out and opens the login page	Automated (frontend)
TC_KB_01	test_knowledge_files: test_wrong_file_types_are_refused, test_zip_that_is_not_a_word_document_is_refused, test_document_without_text_is_refused; test_knowledge_api: test_too_large_file_is_refused	Automated
TC_KB_02	test_knowledge_files: test_pdf_is_recognised_and_read, test_docx_is_recognised_and_read, test_long_text_is_split_with_overlap; test_knowledge_api: test_upload_indexes_the_document	Automated
TC_KB_03	test_knowledge_api: test_failed_indexing_shows_a_plain_message_and_can_be_retried; test_embeddings	Automated
TC_KB_04	test_knowledge_api: test_search_finds_the_closest_passage, test_search_can_be_limited_to_one_collection	Automated
TC_KB_05	test_knowledge_api: test_delete_removes_document_and_passages, test_candidates_cannot_use_the_knowledge_base	Automated
TC_HA_01	test_hiring_assistant: test_new_job_starts_as_draft_profile, test_profile_weights_are_a_suggestion_unless_the_interviewer_gave_them	Automated
TC_HA_02	test_hiring_assistant: test_rag_passages_become_jd_sources, test_rag_search_widens_to_all_collections_when_nothing_found	Automated
TC_HA_03	test_hiring_assistant: test_profile_then_jd_pauses_for_review_then_approve, test_reject_rewrites_with_the_feedback, test_edit_saves_a_new_version_that_still_needs_approval, test_chat_waits_until_the_review_is_done	Automated
TC_HA_04	test_hiring_assistant: test_the_deadline_is_read_as_local_time, test_a_deadline_in_the_past_goes_back_to_the_llm, test_a_new_deadline_updates_the_posted_job	Automated
TC_HA_05	test_hiring_assistant: test_an_off_topic_question_gets_the_fixed_reply_only, test_a_mixed_message_does_the_job_part_and_skips_the_rest; test_assistant_scope (4 tests)	Automated
TC_HA_06	test_ai_models (7 tests); test_llm: test_a_daily_limit_is_not_retried	Automated

Test case ID	Verified by	Type
TC_HA_07	test_hiring_assistant: test_ai_outage_gives_a_clear_503, test_try_again_continues_the_saved_turn, test_long_messages_are_refused, test_a_never_ending_turn_is_stopped_with_a_clear_message	Automated
TC_COL_01	test_hiring_assistant: test_approve_posts_the_job_to_the_board, test_failed_post_keeps_the_jd_and_can_be_tried_again; test_posting (7 tests); jobboard-mcp tests (10)	Automated
TC_COL_02	test_collector: test_a_new_cv_is_collected_once, test_answers_without_job_consent_or_cv_are_skipped, test_the_job_post_link_prefills_the_job, test_one_failing_cv_does_not_stop_the_others	Automated
TC_COL_03	test_collector: test_a_wrong_file_type_is_rejected_and_not_saved, test_a_too_large_cv_is_rejected_without_downloading	Automated
TC_COL_04	test_collector: test_a_cv_sent_after_the_closing_date_is_not_collected, test_no_cvs_by_hand_after_the_closing_date	Automated
TC_COL_05	test_collector: test_interviewers_can_upload_cvs_by_hand, test_an_upload_has_a_file_limit, test_candidates_cannot_upload_cvs	Automated
TC_COL_06	test_collector: test_interviewers_see_the_received_cvs, test_interviewers_can_view_and_remove_an_applicant, test_candidates_cannot_view_or_remove_cvs	Automated
TC_SCR_01	test_screening_flow: test_a_screening_run_ranks_excludes_and_skips, test_screening_needs_a_profile_and_cvs, test_only_one_run_at_a_time	Automated
TC_SCR_02	test_cv_text: test_a_scanned_cv_is_read_with_ocr, test_a_blank_scan_is_unreadable_with_a_clear_reason	Automated
TC_SCR_03	test_screening_rules: test_parsing_finds_the_main_facts, test_overlapping_jobs_count_once, test_more_date_formats_are_read	Automated
TC_SCR_04	test_screening_rules: test_mentions_also_counts_related_and_negation, test_denials_and_wishes_do_not_count_but_versions_do	Automated
TC_SCR_05	test_screening_rules: test_the_gate_excludes_missing_must_haves_and_too_few_years, test_unknown_years_never_exclude	Automated
TC_SCR_06	test_ai_checks (7 tests); test_screening_flow: test_no_dates_and_a_second_look_before_excluding	Automated
TC_SCR_07	Synthetic CV set compared with the answer key (test-data/synthetic-cvs/EXPECTED.md)	Manual
TC_SCO_01	test_screening_rules: test_scores_are_weighted_and_explained	Automated
TC_SCO_02	test_screening_rules: test_weights_must_add_up_to_100; test_screening_flow: test_weights_set_by_the_interviewer_win_and_can_be_reset	Automated
TC_SCO_03	test_screening_flow: test_the_same_cvs_get_the_same_scores_every_time	Automated
TC_SCO_04	test_post_interview: test_the_report_merges_scores_and_waits_for_review	Automated
TC_TAL_01	test_talent_rediscovery: test_the_pool_has_only_people_who_agreed_and_are_free, test_old_applications_and_people_who_already_applied_are_left_out	Automated
TC_TAL_02	test_talent_rediscovery: test_the_check_suggests_past_applicants_who_fit, test_check_again_needs_a_profile_and_one_check_at_a_time	Automated
TC_TAL_03	test_talent_rediscovery: test_adding_a_match_copies_the_application, test_an_upload_joins_the_pool_only_when_the_interviewer_confirms	Automated
TC_TAL_04	test_talent_rediscovery: test_leaving_the_pool_hides_the_suggestion, test_candidates_cannot_see_or_use_the_talent_pool	Automated

Test case ID	Verified by	Type
TC_SCH_01	test_interviews: test_top_n_mentions_a_tie_at_the_cut, test_excluded_and_unknown_names_are_explained, test_shortlist_replace_add_and_remove	Automated
TC_SCH_02	test_interviews: test_invitations_wait_for_the_close_and_a_new_screening	Automated
TC_SCH_03	test_interviews: test_times_are_checked, test_free_slots_skip_weekends_and_booked_times, test_busy_calendar_times_block_a_time, test_an_unreachable_calendar_never_blocks_interviews	Automated
TC_SCH_04	test_interviews: test_confirmed_plan_creates_interviews_and_checks_again, test_cancelling_the_card_saves_nothing, test_shortlist_then_invite_through_the_chat	Automated
TC_SCH_05	test_interviews: test_invited_candidates_get_the_knowledge_base_template, test_a_failed_email_keeps_the_interview, test_html_is_escaped_and_the_link_always_included; email-mcp and calendar-mcp tests	Automated
TC_SCH_06	test_interviews: test_reschedule_and_cancel	Automated
TC_IDV_01	test_identity: test_only_invited_applicants_with_consent_have_their_photo_processed, test_a_photo_is_stored_encrypted_with_a_small_copy, test_photo_types_size_and_rotation_are_handled	Automated
TC_IDV_02	test_identity: test_the_candidate_checks_in_only_when_the_photo_is_ready, test_unusable_pictures_are_refused_with_what_to_change, test_declining_deletes_the_photo_at_once	Automated
TC_IDV_03	test_identity: test_the_match_score_scale, test_the_decision_rules, test_a_doubtful_check_in_never_shows_the_profile, test_another_applicant_at_the_check_in_is_named	Automated
TC_IDV_04	test_identity: test_candidates_never_see_the_result, test_the_photo_is_shown_to_interviewers_only	Automated
TC_IDV_05	test_identity: test_photos_are_deleted_after_the_retention_time, test_a_recent_interview_keeps_its_photo	Automated
TC_VID_01	test_interview_room: test_each_person_gets_a_ticket_for_this_room_only, test_the_candidate_sees_only_their_interview	Automated
TC_VID_02	test_interview_room: test_nobody_joins_before_the_invitation_is_emailed, test_the_candidate_waits_until_let_in, test_nobody_is_let_in_after_the_end	Automated
TC_VID_03	Two-browser call: mute, camera, screen share and end session	Manual
TC_VID_04	Page reload and network drop during a call; 45-minute stability test (#647758)	Manual
TC_VID_05	test_interview_room: test_the_first_join_makes_the_interview_live, test_a_forged_webhook_is_refused, test_ending_closes_the_call_and_the_link	Automated
TC_VID_06	test_interview_room: test_chat_between_interviewer_and_candidate, test_chat_checks_length_speed_and_the_end	Automated
TC_VID_07	test_interview_room: test_cv_panel_and_the_reduced_summary, test_the_candidate_never_sees_the_cv_panel_or_the_assistant	Automated
TC_TRN_01	test_transcriber: test_each_track_is_one_speaker, test_the_transcriber_only_listens, test_live_text_goes_to_the_interviewer_only	Automated
TC_TRN_02	test_transcriber: test_pieces_grow_one_turn_until_a_real_pause, test_silence_and_very_long_answers, test_turn_times_follow_when_the_audio_was_heard	Automated
TC_TRN_03	test_transcriber: test_the_stream_recovers_when_deepgram_drops, test_a_broken_line_loses_no_audio_and_repeats_none	Automated
TC_TRN_04	test_transcriber: test_a_whole_stream_saves_the_turns_and_the_last_words, test_the_last_live_words_are_kept_when_deepgram_is_too_slow	Automated

Test case ID	Verified by	Type
TC_TRN_05	test_transcriber: test_refusals_become_plain_problems, test_a_refused_key_is_shown_and_not_retried, test_a_live_interview_without_the_transcriber_says_so	Automated
TC_TRN_06	test_transcriber: test_the_stream_asks_for_our_settings, test_job_words_help_with_technical_terms	Automated
TC_RAS_01	test_interview_room: test_the_room_assistant_has_its_own_thread_and_saves_cv_questions, test_the_room_assistant_declines_off_topic_questions	Automated
TC_QA_01	test_post_interview: test_ending_an_interview_starts_its_analysis, test_a_failed_analysis_continues_without_repeating_finished_steps, test_only_an_ended_interview_is_analysed	Automated
TC_QA_02	test_qa_pairs: test_units_are_ordered_numbered_and_linked, test_lines_from_the_wrong_person_or_that_do_not_exist_are_dropped, test_only_answered_job_and_company_questions_are_scored	Automated
TC_QA_03	test_post_interview: test_answers_are_evaluated_in_batches; test_qa_pairs: test_an_evaluation_keeps_only_valid_lines_requirements_and_sources	Automated
TC_QA_04	test_post_interview: test_the_analysis_scores_job_and_company_answers_and_maps_requirements	Automated
TC_QA_05	test_qa_pairs: test_coverage_shows_which_units_explored_each_requirement	Automated
TC_QA_06	test_post_interview: test_the_review_checks_cv_claims_and_scores_the_interview	Automated
TC_QA_07	test_assessment: test_the_interview_score_has_six_dimensions_with_evidence, test_without_scored_answers_there_is_no_interview_score	Automated
TC_QA_08	test_interview_quality (5 tests)	Automated
TC_QA_09	test_post_interview: test_the_analysis_scores_job_and_company_answers_and_maps_requirements (names removed); test_ai_checks: test_the_cv_is_sent_without_contact_details_or_the_name	Automated
TC_REP_01	test_post_interview: test_the_report_merges_scores_and_waits_for_review, test_a_report_needs_a_finished_analysis	Automated
TC_REP_02	test_assessment: test_the_position_steps_follow_the_combined_score; test_post_interview: test_without_scored_answers_the_report_says_it_rests_on_the_cv	Automated
TC_REP_03	test_assessment: test_only_existing_evidence_ids_are_kept; test_post_interview: test_unsupported_conclusions_are_repaired_then_marked	Automated
TC_REP_04	test_post_interview: test_changes_requested_then_approved, test_a_change_request_needs_a_comment	Automated
TC_REP_05	test_post_interview: test_the_export_is_a_printable_html_file; test_assessment: test_a_draft_is_marked_on_every_page_and_an_approval_is_named	Automated
TC_REP_06	test_post_interview: test_the_reports_of_a_job_list_every_ended_interview; reportFormat.test.js	Automated
TC_REP_07	test_post_interview: test_candidates_never_see_analyses_or_reports; source panel checked in the UI	Automated + Manual
TC_SEL_01	test_final_ranking (5 tests); test_selection_email: test_the_reports_list_ranks_only_approved_reports	Automated
TC_SEL_02	test_selection_email: test_only_after_the_report_is_approved, test_the_checked_email_is_sent_once, test_a_failed_email_does_not_mark_the_candidate, test_candidates_cannot_see_or_send_it	Automated

Test case ID	Verified by	Type
TC_DM_01	test_privacy: test_deleting_a_candidate_removes_everything_about_them; test_models: test_candidate_data_is_deleted_with_the_candidate	Automated
TC_DM_02	test_job_deletion (7 tests)	Automated
TC_DM_03	test_dashboard (8 tests)	Automated
TC_DM_04	test_privacy: test_only_state_that_is_still_needed_is_kept	Automated
TC_NFR_01	Timed screening of 20–22 synthetic CVs (VERIFICATION.md §9.3)	Manual measurement
TC_NFR_02	Live transcript timing during the 45-minute test (VERIFICATION.md §9.3)	Manual measurement
TC_NFR_03	API probe during a 29-CV screening, 1 call per second for 3 minutes	Manual measurement
TC_NFR_04	test_runner: test_successful_run_is_completed, test_progress_is_kept_when_a_run_fails	Automated
TC_NFR_05	45-minute stability test with background screening (#647758)	Manual
TC_NFR_06	Report time after the 45-minute test (VERIFICATION.md §9.3)	Manual measurement
TC_NFR_07	test_post_interview: test_answers_are_evaluated_in_batches; request count in the README “AI usage” section	Automated + Review
TC_SAF_01	Walk-through of each confirmation step (JD, posting, invitations, report, selection email)	Manual
TC_SAF_02	email-mcp tests (test recipient override)	Automated
TC_SEC_01	test_security: test_password_is_stored_as_a_hash_and_can_be_checked, test_unknown_user_never_matches	Automated
TC_SEC_02	test_security: test_expired_token_is_rejected, test_token_signed_with_another_secret_is_rejected, test_token_with_unknown_role_is_rejected	Automated
TC_SEC_03	test_security: test_link_token_hash_is_fixed_length_and_hides_the_token; test_auth_api: test_a_forwarded_link_without_the_code_gets_nowhere	Automated
TC_SEC_04	test_knowledge_files, test_collector and test_identity upload checks; code review of request models	Automated + Review
TC_SEC_05	Repository scan for keys and personal data; .gitignore review (VERIFICATION.md §9.1)	Review
TC_SEC_06	test_runner: test_unexpected_error_hides_technical_details; test_llm: test_api_returns_503_when_ai_fails; client.test.js	Automated
TC_SEC_07	Response header check on the running stack	Manual
TC_SEC_08	Dependency and image pin review (VERIFICATION.md §9.1)	Review
TC_SQA_01	test_runner: test_interrupted_runs_resume_at_startup; test_checkpointer: test_graph_resumes_after_restart	Automated
TC_SQA_02	Ruff check and code review	Review
TC_SQA_03	Full test run: backend, MCP servers and frontend	Automated
TC_SQA_04	End-to-end walk-through with synthetic data	Manual
TC_SQA_05	Fresh clone started with the documented commands only	Manual
TC_SQA_06	test_models: test_no_protected_attribute_columns; prompt review	Automated + Review
TC_UI_01	Screens checked at 375, 768 and 1280 px in light and dark theme	Manual
TC_OR_01	README “AI provider and data sent to it” and “Privacy note” review	Review

Test case ID	Verified by	Type
TC_OR_02	README review; /docs OpenAPI page opened	Review
TC_OR_03	Git tag and /docs contents checked before submission	Review